

Object Detection System Report using DETR

Methodology Description

This report outlines the implementation of an object detection system utilizing the DETR model. The system was trained on a dataset formatted for object detection, specifically the COCO dataset, which comprises images paired with annotations such as bounding boxes and category labels.

Implementation Details

Key Code Components

1. **Collate Function:** This function prepares batches of images alongside their corresponding target annotations for training. It stacks images into a tensor and organizes bounding boxes and class labels into a suitable format for the model.
2. **Training and Validation Functions:**
 - **train_epoch:** This function manages the training of the model for one epoch, calculates the loss, and updates model parameters.
 - **validate_model:** Evaluates the model on the validation dataset and computes the average validation loss.
3. **Training Loop:** The **train_model** function oversees the overall training process, including logging loss metrics to TensorBoard and saving the best model based on validation loss.
4. **Result Checking Function:** The **check_model_results** function visualizes predictions for individual images from the test dataset. It filters predictions based on a confidence threshold and displays both the ground truth and the predicted bounding boxes.
5. To optimize calculations, all images were resized to a consistent dimension, which ensured uniform input for the model. Additionally, images were batched together during processing to enhance computational efficiency, allowing for faster training and inference by leveraging parallel processing capabilities.

Data Augmentation

To enhance the model's performance, data augmentation techniques were employed, effectively tripling the size of the training dataset. The following transformations were applied:

```
transform = A.Compose([ A.HorizontalFlip(p=0.5), A.VerticalFlip(p=0.5),  
A.RandomBrightnessContrast(p=0.3), A.ShiftScaleRotate(shift_limit=0.15, scale_limit=0.15,  
rotate_limit=45, p=0.5), ], bbox_params=A.BboxParams(format='coco',  
label_fields=['category_ids']))
```

Enhanced Dataset: The training dataset was effectively increased threefold through augmentation techniques such as:

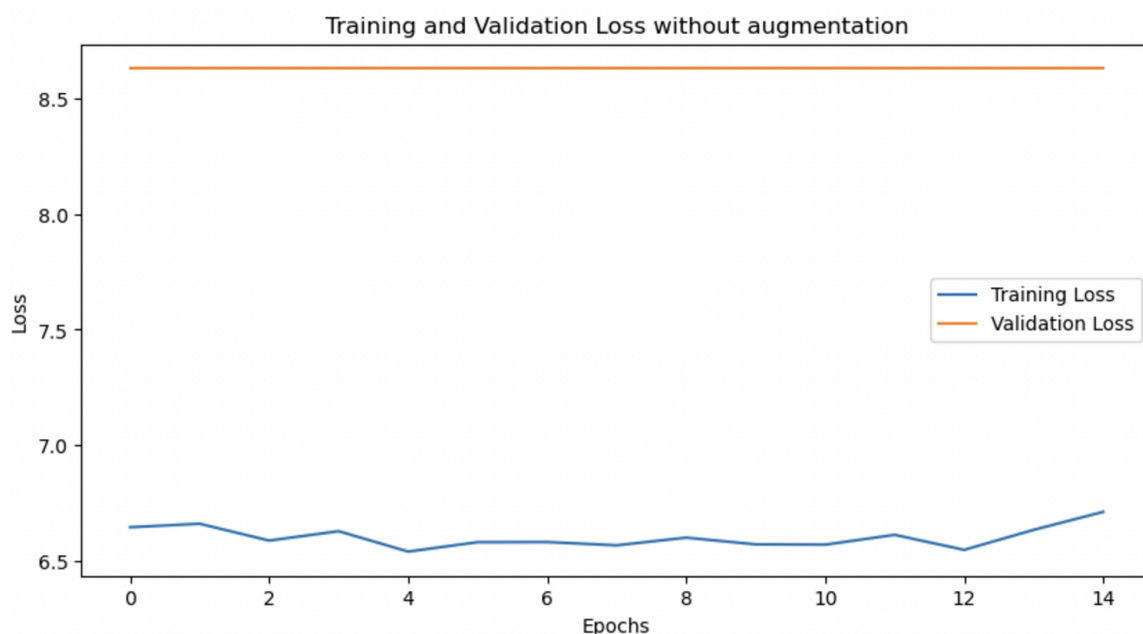
- Horizontal and vertical flips
- Random brightness and contrast adjustments
- Shift, scale, and rotation transformations

Results Analysis

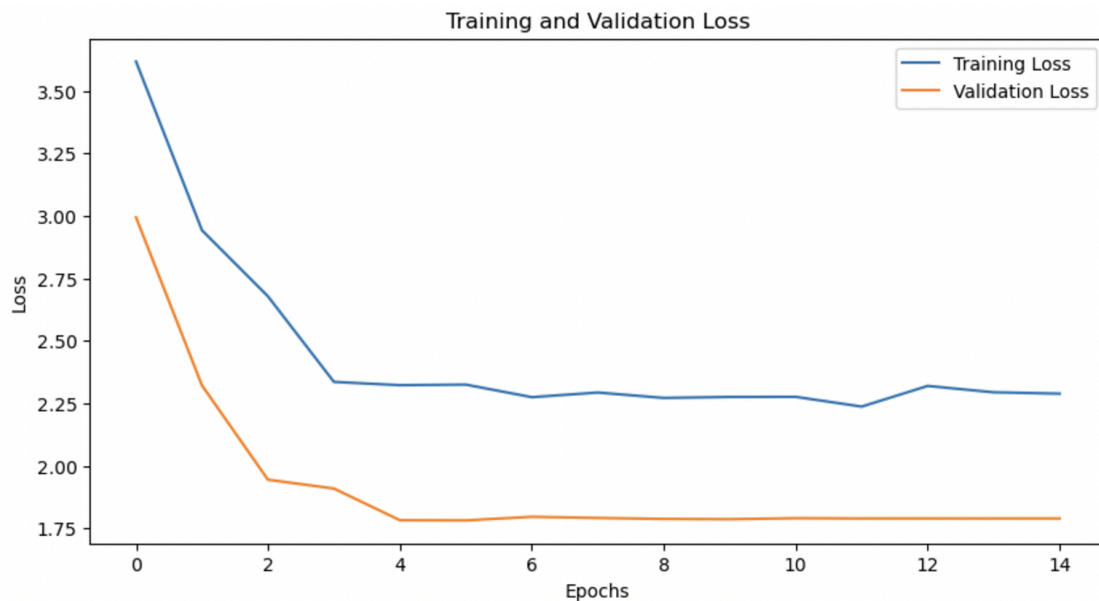
Training and Validation Loss

The training and validation loss metrics were closely monitored throughout the training process. With the augmented dataset, the model exhibited a more pronounced decrease in both training and validation loss compared to training without augmentation.

- **Without Augmentation:** The model struggled to converge effectively, leading to relatively high validation loss.



- **With Augmentation:** The training loss decreased significantly, reaching a lower plateau, while the validation loss also showed improvement, indicating better generalization.



Average Precision (AP) Score

The model trained on the augmented dataset achieved an impressive Average Precision (AP) score of **0.771** at an IoU threshold of 0.5. This score reflects a substantial improvement over the baseline model trained without augmentation, showcasing the effectiveness of the applied data augmentation techniques.

Discussion of Challenges and Solutions

Challenges

The DETR model initially struggled with poor performance on smaller datasets, which limited its ability to detect objects effectively. Additionally, there were challenges in correctly applying data augmentation techniques while ensuring that the annotations remained accurate. This required careful handling to maintain the integrity of the bounding box information during transformations.

Solutions

To address this issue:

- **Data Augmentation:** As previously mentioned, I implemented various data augmentation techniques, which enhanced the diversity of the training dataset and significantly increased its size. This strategy helped improve the model's ability to recognize balloons by providing a broader range of examples.

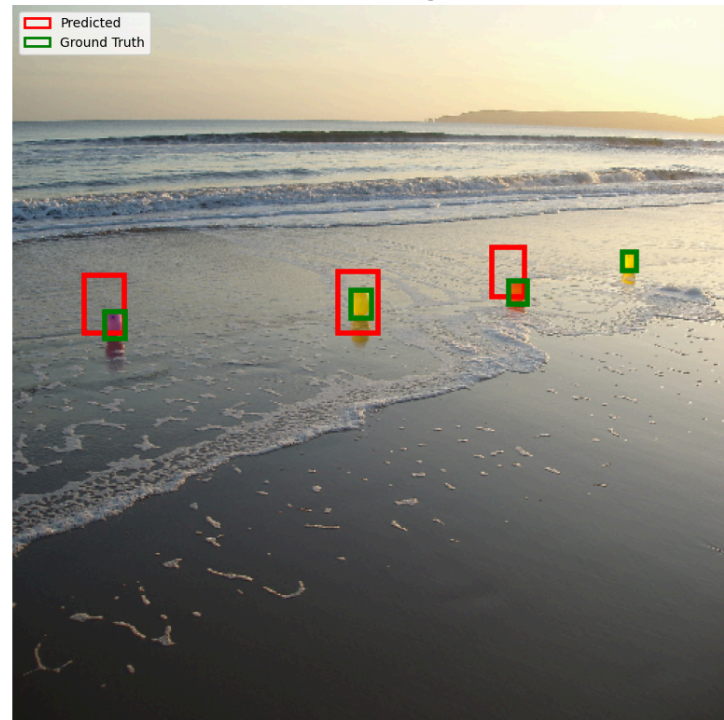
Visualizations

To illustrate the model's performance, the entire dataset of images was processed using the DETR model to visualize the predicted bounding boxes alongside the ground truth. Below are examples where the model, trained on the augmented training dataset, successfully predicted bounding boxes:

Model Predictions with augmentation

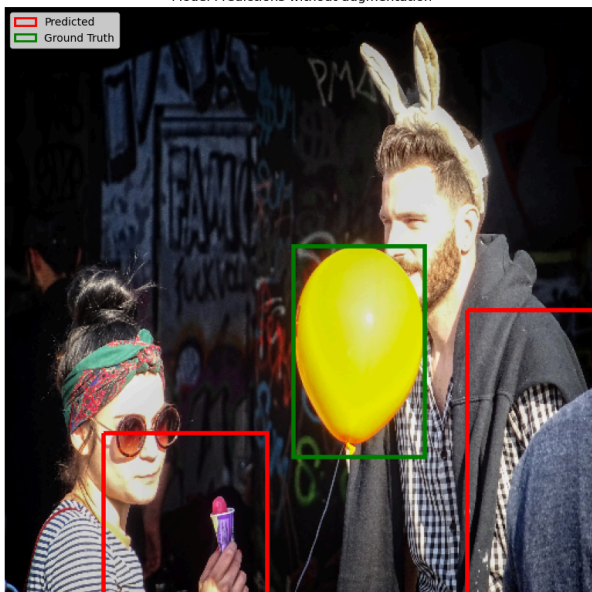


Model Predictions with augmentation



And here are the results for the model trained without data augmentation:

Model Predictions without augmentation



These visualizations provide a clear representation of the model's performance, highlighting both successful detections and areas for improvement.

Conclusion

The implementation of the DETR model for object detection demonstrated significant improvements when utilizing data augmentation techniques. The enhancement in Average Precision and the reduction in loss metrics underline the effectiveness of these strategies in addressing initial challenges related to object size detection.